THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1013/1113: ENGINEEIRNG MECHANICS AND MATERIALS

12 December, 2013 TIME: 14:30 – 17:30 (3 hours)

Answer **5 (FIVE)** questions

All questions carry equal marks

Use a separate answer book for each section

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

1. Figures 1(a) to 1 (c) show a simply supported beam ABC with an overhang subject to different loads. Draw the shear force and bending moment diagrams, at each of the following loading conditions:

   (a) When the beam ABC is subject to a uniformly distributed load, w (kN/m) applied along the whole length of beam as shown in Figure 1(a) **[8 Marks]**

   (b) When the beam ABC is subject to a point load, P (kN) applied at free-end of the overhang as shown in Figure 1(b). **[6 Marks]**

   (c) When the beam ABC is subject to a moment, M (kNm) applied at point B as shown in Figure 1(c). **[6 Marks]**

Note:
- All bending moment diagrams are to be drawn on the tension side
- Position and magnitude of maximum bending moment to be shown on all bending moment diagrams

2. Figure 2 shows a symmetrical tee-section beam has flange and web lengths of 150 and 90mm, respectively. The flange and web thickness is uniform at 10mm. The beam is simply-supported of span 1.5m subject to a uniformly distributed load, w (kN/m).

   (a) Evaluate the centroid, h (in mm) of the tee-section measured from the extreme fibre of the top flange. **[4 marks]**

   (b) Calculate the uniformly-distributed load w (in kN/m) which can be applied if the maximum tensile stress is not to exceed 100MPa. **[10 marks]**

   (c) What is the greatest bending stress (in MPa) in the flange? **[6 marks]**

3. Figure 3 shows two steel struts AB and BC are connected by a hinge at point B. A vertical force F is applied at point B. Both the struts have identical L-shaped cross-section with a constant thickness of 7 mm and a modulus of elasticity, E of 200GPa.

   (a) By comparing the length and the proportion of forces between member AB and member BC, show that member AB is the critical element in this structure. **[8 marks]**

   (b) Based on the Euler's buckling theory, determine the magnitude of force F (in kN). **[12 marks]**

FIGURES

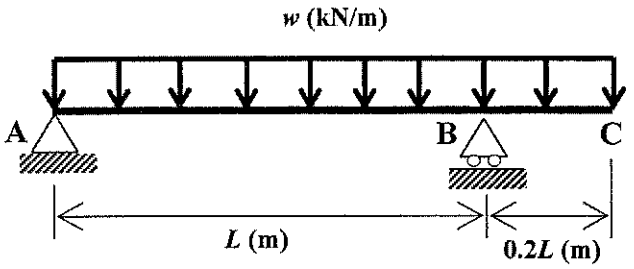


Figure 1(a) (Not to scale)

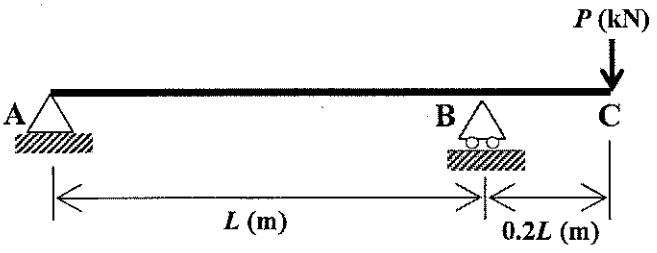


Figure 1(b) (Not to scale)

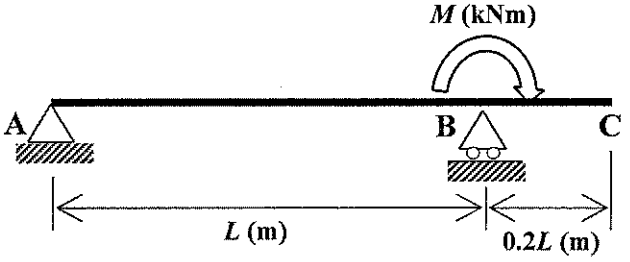


Figure 1(c) (Not to scale)

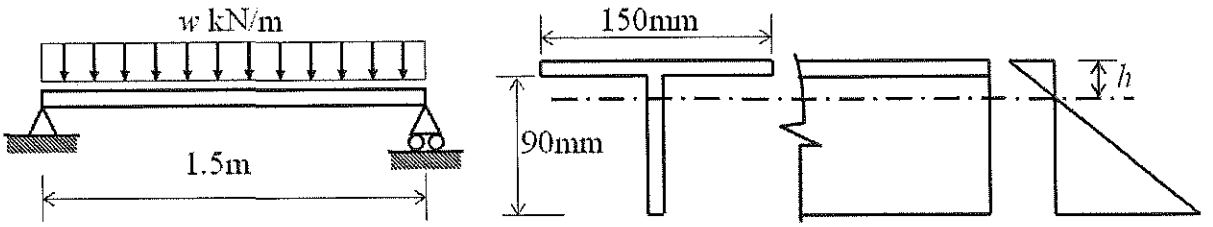


Figure 2 (Not to scale)

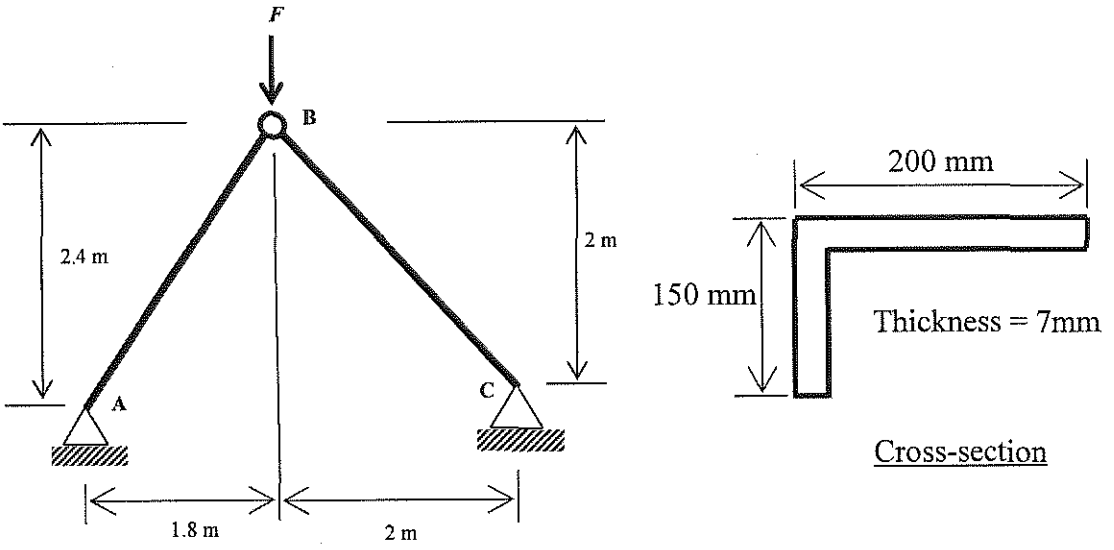


Figure 3 (Not to scale)

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